

IB 451 - Fall 2025 - Final Exam

Select 8 of the following 10 questions to answer.

Answers should be around 1-2 paragraphs each.

Submit your answers on Canvas by 12/18 at 11:59pm CST.

1. In the beloved program Sesame Street, Cookie Monster, upon encountering a plate of cookies, will always eat all of the cookies present. While this makes for enjoyable puppetry, this does not necessarily align with the principles of the optimal foraging theory. Explain the general assumptions that optimal foraging theory makes. Which of these assumptions might not align with Cookie Monster's behavior?
2. Outline your own lecture on a topic/subfield within animal behavior that was not the focus of a class lecture. A full answer for this question would include.
 - a. Identify the topic/subfield of interest (if you are not sure what counts as a topic/subfield, you can ask us).
 - b. 3 primary learning objectives.
 - c. 2 case studies/papers to cover.
 - d. 1 class discussion question/activity.
3. A friend of yours asks why you took this class when animals all behave the same way (all animals eat, all animals reproduce, all animals try not to get eaten). It is your task to explain to your friend how this isn't true, using 3 examples: 1) within an individual organism, 2) within a species, and 3) across species.
4. A construction company hires you (a famed behavioral ecologist) as a consultant for their dam project. During a Q&A, a local politician questions whether your skills are necessary for what is ultimately a construction project. How would you justify your inclusion on this project?
5. You are an animal living in a stressful environment and must do everything you can to survive. When encountering another member of your species, you can either work together to obtain food or you can cheat your peer. Provide what information you would need to know about yourself, the interaction, and the environment before deciding which strategy to adopt.

6. During this class, we've talked a lot about group living. Meerkats are one such organism that engage in group living and cooperation, and generally fare better for it - however, they will still form distinct groups within a population. Why do all meerkats (in a population) not simply join up into one super group?
7. There are a great many creatures in literature, video games, and television (e.g., dragons, Pokemon, sand worms). Most of these stories do not delve into the details of their creatures' life history. For a fictional creature of your choice, explain the mating system and parental care strategy you would expect it to have based on environment, lifespan, and life history
8. One concept we've talked about in this class is behavioral plasticity. However, being plastic often incurs costs (e.g., higher maintenance). When is it beneficial for an animal to simply have an evolved and fixed behavior, and when is it beneficial to display behavioral plasticity?
9. Cognitive ability is often touted as an evolutionary adaptation that allows animals to respond to different environments. Why, then, do some of the world's most successful animals not display high levels of cognitive ability (e.g. flies, jellyfish, nematodes, beetles)?
10. Being able to communicate confers many benefits (e.g., increased foraging success, better quality mates). In theory, a more complex signal can communicate more information. Why do we not see highly complex signals for all organisms? Why, even in species with complex signals (e.g. highly complex birdsong) do we see low-complexity signals?